

1. What does “scope” mean in Python?

- A. The size of a variable
- B. The region where a variable can be accessed
- C. The data type of a variable
- D. The lifetime of a program

Ans: **B.** The region where a variable can be accessed

2. Which rule does Python use to find a variable?

- A. FIFO
- B. LIFO
- C. LEGB
- D. FILO

Ans: **C.** LEGB

3. What does LEGB stand for?

- A. Local, External, Global, Built-in
- B. Local, Enclosing, Global, Built-in
- C. Local, Enclosing, General, Basic
- D. Local, Global, Enclosing, Built-in

Ans: **B.** Local, Enclosing, Global, Built-in

4. A variable created inside a function normally has what scope?

- A. Global
- B. Built-in
- C. Local
- D. Enclosing

Ans: **C.** Local

5. What is the output?

```
x = 10
```

```
def test():
```

```
    x = 20
```

```
    print(x)
```

```
test()
```

A. 10

B. 20

C. 30

D. Error

Ans: **B.** 20

6. What is the output?

```
x = 10
```

```
def test():
```

```
    print(x)
```

```
test()
```

A. 0

B. 10

C. None

D. Error

Ans: **B.** 10

7. Which keyword is used to modify a global variable inside a function?

A. global

B. extern

C. public

D. outer

Ans: A. global

8. What is the output?

x = 10

```
def change():
```

```
    global x
```

```
    x = 20
```

```
change()
```

```
print(x)
```

A. 10

B. 20

C. None

D. Error

Ans: B.20

9. What happens here?

```
x = 10
```

```
def change():
```

```
    x = x + 5
```

```
    print(x)
```

```
change()
```

A. Prints 15

B. Prints 10

C. Raises an error

D. Prints 5

Ans: C. Raises an error

10. Which keyword allows a nested function to modify a variable in its enclosing function?

A. global

B. local

C. nonlocal

D. outer

Ans: C. nonlocal

11. What is the output?

```
def outer():
```

```
    x = 10
```

```
    def inner():
```

```
        print(x)
```

```
    inner()
```

```
outer()
```

A. 0

B. 10

C. Error

D. None

Ans: B. 10

12. What is the output?

```
x = "global"
```

```
def outer():
```

```
    x = "enclosing"
```

```
def inner():  
    print(x)
```

```
inner()
```

```
outer()
```

A. global

B. enclosing

C. local

D. Error

Ans: B. enclosing

13. Which scope is searched first according to LEGB?

A. Global

B. Built-in

C. Enclosing

D. Local

Ans: D. Local

14. Which scope is searched last according to LEGB?

A. Local

B. Enclosing

C. Global

D. Built-in

Ans: D. Built-in

15. What is the output?

```
x = 100
```

```
def fun():
```

```
    x = 50
```

```
    print(x)
```

```
fun()
```

```
print(x)
```

A. 50, 50

B. 100, 100

C. 50, 100

D. 100, 50

Ans: C. 50,100

16. What is the output?

```
def fun():
```

```
    y = 20
```

```
fun()
```

```
print(y)
```

A. 20

B. 0

C. None

D. NameError

Ans: D. NameError

17. Which variable is accessible throughout a module, unless shadowed?

A. Local variable

B. Global variable

C. Enclosing variable

D. Temporary variable

Ans: B. Global variable

18. What is the output?

`x = 5`

```
def fun():
```

```
    print(x)
```

```
    x = 10
```

```
fun()
```

A. 5

B. 10

C. UnboundLocalError

D. NameError

Ans: C. UnboundLocalError

19. Why does Question 18 produce an error?

- A. x cannot be printed
- B. Assignment makes x local to the function
- C. Global variables cannot be used in functions
- D. Python does not support reassignment

Ans: B. Assignment makes x local to the function

20. What is the output?

```
x = 10
```

```
def fun():
```

```
    global x
```

```
    x += 5
```

```
fun()
```

```
print(x)
```

- A. 10
- B. 15
- C. 5
- D. Error

Ans: B. 15

21. What is the output?

```
def outer():
```

```
    x = 10
```

```
    def inner():
```

```
        nonlocal x
```

```
        x = 20
```

```
    inner()
```

```
    print(x)
```

```
outer()
```

A. 10

B. 20

C. Error

D. None

Ans: B. 20

22. What does nonlocal refer to?

A. A variable in the global scope

B. A variable in the built-in scope

C. A variable in an enclosing function scope

D. A variable outside Python

Ans: C. A variable in an enclosing function scope

23. Which of the following is an example of a built-in name?

A. my_variable

B. x

C. len

D. total

Ans: C. len

24. What is the output?

```
def fun():  
    print(len([1, 2, 3]))
```

fun()

A. 2

B. 3

C. 1

D. Error

Ans: B.3

25. What is variable shadowing?

A. Deleting a variable

B. Creating a variable with the same name in a narrower scope

C. Making a variable global

D. Copying a variable

Ans: B. Creating a variable with the same name in a narrower scope

26. What is the output?

```
x = 1
```

```
def outer():
```

```
    x = 2
```

```
    def inner():
```

```
        x = 3
```

```
        print(x)
```

```
    inner()
```

```
outer()
```

A. 1

B. 2

C. 3

D. Error

Ans: C.3

27. What is the output?

```
x = 1
```

```
def outer():
```

```
    x = 2
```

```
def inner():
```

```
    print(x)
```

```
inner()
```

```
outer()
```

A. 1

B. 2

C. Error

D. None

Ans: B.2

28. Which statement about local variables is TRUE?

A. They are always accessible outside the function

B. They are normally accessible only within the function where they are defined

C. They are always global

D. They can only contain numbers

Ans: B. They are normally accessible only within the function where they are defined

29. What is the output?

```
x = 10
```

```
def outer():
```

```
x = 20
```

```
def inner():  
    nonlocal x  
    x += 5
```

```
inner()  
print(x)
```

```
outer()
```

A. 10

B. 20

C. 25

D. Error

Ans: C. 25

30. What will Python do if a name is not found in Local, Enclosing, Global, or Built-in scope?

A. Create the variable automatically

B. Return None

C. Raise a NameError

D. Search the internet

Ans: C. Raise a NameError

